Harpuneet Singh

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Education

University of Toronto, *HBSc Computer Science (Co-op)* Aug 2023 – Dec 2028
Recognitions: Dr. Babak Farzad's Award of Leadership, Laidlaw Research Scholar, PRISM Fellow, Dean's List Scholar

Technical Skills

Languages: C++, Python, C, Verilog, MicroPython, Assembly, R, MATLAB, Java, MySQL, HTML, JavaScript, Kotlin, Git
Prototyping: Arduino, Raspberry Pi, IoT Sensors, AdaFruit, SolidWorks, AutoCAD, Fusion, EasyEDA, Altium, Logisim
Equipment: Fluorescence microscope, Optical spectrometer, Microscope cameras, Tunable filters, Dichroics, Centrifuge, Gel electrophoresis, Thermocycler, Piezo stages, Oscilloscope, Network analyzer, FPGAs, Reflow oven, 3D printers
Network Interfaces: LoRaWAN, Wi-Fi, Bluetooth LE, NFC/RFID, USB, Ethernet, MQTT, I^2C, SPI, UART, CAN

Experience

SpectraSip, *Interactive Intelligent Systems Laboratory (Tokyo, Japan)* May 2026 – Present

- Developing a low-cost smart coaster to identify near infrared spectral fingerprints of adulterated drinks
- Ran **180+ controlled trials** across varying drug concentrations, drink compositions, and lighting conditions
- Trained an NIR spectral classifier using multiband spectral signatures, achieving **92%** adulterant detection accuracy
- Deployed the system in Tokyo nightlife venues & cut ambient-light interference by 75% across 4 hardware iterations

Opto-Dye, *Wheeler Microfluidics Laboratory (Toronto, Canada)* Mar 2025 – Apr 2026

- Designed a microfluidic–optical system enabling low-resource forensic hyperspectral microscopy across 350-1080 nm
- Developed a deep-learning cell classifier using **Detectron2** and **μSAM** for unstained multi-channel microscopy samples
- Integrated the imaging platform with a microfluidic DNA-analysis chip, reducing sample processing time by **70%**
- Validated performance on 200+ samples & achieved **97% accuracy** against traditional staining and hyperspectral methods
- **Publication:** *Singh H, Verma D and Mariakakis A.* Multispectral LED microscopy for rapid, dye-free sperm cell identification in forensic samples F1000Research 2026 [poster]

ChromaFlash, *Computational Health & Interaction Laboratory (Toronto, Canada)* Apr 2024 – Dec 2025

- Built an adaptive illumination system to enhance optical contrast between materials via a programmable LED flash
- Spectrally characterized over 50 materials across 400-900 nm using an optical spectrometer to get absorption features
- Constructed high-speed PCBs using **Altium** to support 7 ns LED switching speed for precise illumination
- Implemented per-channel brightness and timing control on a Raspberry Pi Pico 2 using **C++** and **MicroPython**
- Produced a **13 dB contrast** between materials with a 10 nm illumination granularity and 800 lumens scene brightness

Cognitive Restructuring, *Intelligent Adaptive Interventions Laboratory (Toronto, Canada)* Aug 2024 – Aug 2025

- Developed an LLM-based framework for cognitive restructuring using LangChain, Keras and TensorFlow in **Python**
- Optimized model architecture for a **40% increase** in inference efficiency and a **30% decrease** in processing latency
- Redesigned a more user-friendly UI using React and Node.js in **Javascript**, increasing the daily user base by **200 users**

AgroArmor, *United Nations Innovation Laboratory (Rome, Italy)* Apr 2024 – Nov 2024

- Designed a scalable, modular biogas plant addressing energy production in rural agricultural settings
- Engineered custom PCBs and IoT sensor modules using **Altium** to implement continuous low-power slurry agitation
- Prototyped hardware using **Fusion 360** and **AutoCAD**, modeled gas flow and power consumption using **MATLAB**
- Integrated eco-friendly materials, reducing costs by **30%** while ensuring durability and repair-ability

Mersivity: Inclusive Immersive Tech, *Mannlabs (Toronto, Canada)* Oct 2023 – Jan 2024

- Engineered Virtual Reality(VR) tech for spinal muscular atrophy patients using an EEG-based direct brain interface
- Built 3 Sequential Wave Imaging Devices capable of creating a VR environment from a singular LED strip
- Prototyped with a custom PCB made using **EasyEDA** with addressable LED strips controlled using an ESP32 Feather
- Integrated with a Muse InteraXon EEG headband, using **Python** and **C++** to convert sensor readings into VR actions

Clubs and Extracurriculars

UofTHacks, *Sponsorship Executive* May 2025 – Present
Leading sponsor outreach for UofT's flagship hackathon. Contacted 100+ sponsors, pitched initiatives, and secured funding
USRP (Undergraduate Student Research Program), *Research Mentor* Jan 2025 – Present
Organizing workshops and mentoring 10 undergrads in starting projects, building skills, and navigating opportunities